

RYAN KIM

New Jersey | 201-245-8979 | rkydin404@gmail.com | [linkedin.com/in/ryandki](https://www.linkedin.com/in/ryandki) | [rkydin.github.io](https://github.com/rkydin) | U.S. Citizen

Open to relocation and nationwide travel opportunities

TECHNICAL SKILLS

Mechanical Design: SolidWorks; SolidWorks Simulation; Fusion 360; FreeCAD; Abaqus; Siemens NX; ASME Y14.5-2018; Feature Control Frames; Tolerance Stack-Ups; DFM; DFA; FEA; 2D Drawings; BOM Management

Manufacturing: Prototype Assembly; Fixture Design; Root Cause Analysis; ISO 9001; Documentation Control (Microsoft 365 Suite)

Laboratory: Mechanical Testing; Failure Analysis; Data Analysis; Instron Testing; Optical Microscopy; GLP; Oscilloscopes; DMM

Programming: Python; MATLAB; C++

EXPERIENCE

Volta Space Technologies

April 2026 – July 2026

Mechanical Design Intern

Broomfield, CO

- Designed and built a **laser assembly enclosure** from aluminum extrusions and sheet metal in **SolidWorks** for delivery to the European Space Agency.
- Assembled electromechanical hardware, including **soldering** and **cable harnessing**.
- Designed and **3D-printed optical-mount fixtures** for fibers, lasers, lenses, and mirrors in **SolidWorks** and **Fusion 360**, applying GD&T tolerancing for optical alignment ahead of future TVAC testing.
- Sourced vendor quotes and coordinated manufacturing of aluminum sheet-metal and machined parts, designed and ordered **optical breadboards**, and documented procurement using the **Microsoft 365 suite**.

Robomechanics

Nov 2025 – Feb 2026

Mechanical Engineer (Contract, 3 months)

Teterboro, NJ

- Reduced part count **15%** on pallet-handling robotics assemblies by redesigning and merging components during metric-to-inch CAD conversion, cutting fabrication and assembly labor.
- Converted legacy metric CAD assemblies into inch-based production models and **ASME Y14.5-2018**-compliant drawings within an **ISO 9001** quality program, documented using the **Microsoft 365 suite**.
- Resolved fit-up, interference, and failure issues on sheet-metal/welded assemblies via **tolerance stack-ups** and geometry revisions with machinists and fabricators.
- Built on-demand prototypes directly from rough concept sketches, translating informal design intent into functional hardware under tight turnaround.

Lamont-Doherty Earth Observatory

April 2021 – Aug 2021

Research Assistant

Palisades, NY

- Supported research on microplastic contamination in consumer detergents through sample preparation, filtration, **UV-Vis spectrophotometry**, and cleanroom laboratory testing.

EDUCATION

Northeastern University

Boston, MA

M.S. Mechanical Engineering, Concentration in Materials, Incoming Fall 2026

Sep 2026 – May 2028

Rutgers University, School of Engineering

New Brunswick, NJ

B.S. Biomedical Engineering

Sep 2021 – May 2025

- Coursework in biomaterials, biomechanics, and biomedical devices covering the mechanical, biological, and chemical behavior of engineered systems.
- Performed mechanical testing, failure analysis, and data analysis using **Instron** machines and optical microscopy.
- Completed electronics labs with **oscilloscopes** and **DMMs**, and FEA/simulation coursework in **SolidWorks** and **Abaqus**.

PROJECTS

View full project portfolio: [rkydin.github.io](https://github.com/rkydin)

UltraLight Sleeping Pad Pump

Nov 2025 – Present

Personal Project

- Designed and iterated a backpacking pump through **three hardware revisions** in SolidWorks, engineering a sealed three-piece PETG housing with a centrifugal impeller to hit **~31 g** under a sub-100 g ultralight target.
- Designed a custom **2-layer PCBA in KiCad** for USB-C power input, overcurrent/motor-kickback protection, and power filtering, replacing the off-the-shelf blower and trigger board.